ACI Worldwide Inc.

Transaction Based Pricing

Implementation Guide

BASE24-atm, pos, teller Release 5.0, 5.1, 5.3, 6.0
BASE24-telebanking,billpay 1.1, 1.3, 6.0

October 28, 2005

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ACI Worldwide Inc.
330 South 108th Avenue
Omaha, Nebraska 68154
(402) 390-7600

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Publications Number: BA-UT998-01

Seventh Edition	October 2005/Version 9.0 of TBP Reports
Sixth Edition	June 2002/Version 6.0
Fifth Edition	October 2001/Version 5.0
Fourth Edition	October 2000/Version 4.0
Third Edition	December 1998/Version 3.0
Second Edition	May 1998/Version 2.0
First Edition	April 1996/Version 1.0

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Manual Summary

This manual is organized into the following sections and appendixes. Each section focuses on specific items that must be reviewed or performed by the BASE24 user's release upgrade/conversion team. The appendixes provide supplemental information to assist with this upgrade/implementation.

Section 1 Overview gives a brief and general description of the functionality of the Transaction Based Pricing (TBP) programs.

Section 2 Configuration describes the configuration requirements for the LCONF and NEFS.

Section 3 MTAF lists the layout of the structure of the Monthly Totals Accumulation File (MTAF) .

Section 4 RBAF lists the layout structure of the Remote Banking Accumulation File (RBAF)

Section 5 Reports describes the reports produced when the programs are run.

Appendix A Log Message Format

Appendix B RIP Process Messages

Appendix C Daily Report Program Messages

Appendix D Monthly Report Program Messages

Related Documentation

- BASE24 Logical Network Configuration File Manual
- BASE24 XPNET Network Environment Source File Manual

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1 Overview

This document provides the information for implementing and running of Transaction Based Pricing (TBP) for a Release 5.0/5.1.1/5.1.2/5.3/6.0 BASE24 atm, pos or teller and /or a 1.1/1.3/6.0 BASE24-telebanking/billpay system. It is the property of Applied Communications Inc. (ACI) and the information contained herein is confidential. This document, either in whole or in part, must not be reproduced or disclosed to others or used for purposes other than that for which it has been supplied, without ACI's prior written permission, or, if any part hereof is furnished by virtue of a contract with a third party, as expressly authorized under that contract.

Transaction Based Pricing is a pricing policy tool for the licensing of BASE24 software. It is designed to automatically monitor the transaction volume at a customer site and create daily and monthly transaction volume reports .

The tool consists of three or more programs, dependent on the BASE24 products installed at the customer site. The three core programs are:

- **Report Initiator Program**

  This program, RIP, runs as a BASE24 satellite process. It is responsible for starting daily and monthly report programs as well as creating and purging Monthly Totals Accumulation Files (MTAFs) and/or Remote Banking Accumulation Files (RBAFs).

- **Daily Transaction Volume Report Program**

  These programs analyze the day's Transaction Log File (TLF, PTLF, TTLF or ITLF) and write a record containing transaction volume totals to the MTAF and /or RBAF. The MTAF and/or RBAF is dependent on the BASE24 products installed at a customer site. You can print a report of the totals, if required.

- **Monthly Transaction Volume Report Program**

  These programs analyze the MTAF and/or RBAF at the end of every month and produce a report of the monthly transaction volumes.

The number of daily report programs required for a customer site is dependent on the BASE24 products installed; for example, BASE24-atm, BASE24-pos, BASE24-teller or BASE24-telebanking/billpay. No operator is required to run the daily report programs unless the programs are rerun manually.

The TBP tool reads the entire log file and categorizes the transactions appropriately for tracking monthly volumes.

2 Configuration

This section describes the configuration files which must be set up or verified so that the TBP report programs can function properly.

Section 2.1 lists the files delivered by ACI for Transaction Based Pricing.

Section 2.2 describes the LCONF assigns and params which are required to configure the files correctly.

Section 2.3 describes the NEFS process declaration.

Section 2.4 describes the procedure to run the TBP analysis tool.

Section 2.5 describes how to accelerate TBP objects.

2.1 Object and Template Files

Listed below are the object and template file names and descriptions for Transaction Based Pricing:

ATDRPT	Daily ATM report program
BAMRPT	Monthly report program
MTTPLT	MTAF and RBAF template file (input to FUP)
PSDRPT	Daily POS report program
RBDRPTLO	Daily Remote Banking report program
RBMRPTLO	Remote Banking report program
RIP	Report initiator program
TRDRPT	Daily Teller report program

2.2 Logical Network Configuration File (LCONF)

The Logical Network Configuration File (LCONF) is the central location of file information and parameters for the BASE24 Logical Network. One LCONF exists for each logical network.

The LCONF contains assigns which link the physical names of objects in the logical network to names used by BASE24 code and params which define processing values at the logical network level.

The LCONF assigns and params listed below must be present for the implementation of TBP.

For more details, refer to the ***BASE24 Logical Network Configuration File Manual***.

2.2.1 LCONF Assigns

The LMCF and RPT-ERROR-FILE assigns are already present in the LCONF and are necessary for the TBP analysis tool to run successfully.

There is one copy of this set of the LCONF assigns for each BASE24 product on the customer site that is being monitored. **The assign names are preceded by the product name; for example, ATM-, POS-, TLR-, or TB-.**

- **DAILY-REPORT-DESTINATION**

  The fully-qualified file name of the spooler location for the report produced by the Daily Transaction Volume Report Program. The format is #xxDmmdd, where 'xx' specifies the type of BASE24 product (AT, PS , TR, or RB for BASE24-atm, BASE24-pos, BASE24-teller, or BASE24-telebanking/billpay respectively). If this assign is missing, the daily report will be written to EMS.

  REQUIRED ASSIGN: No

- **DAILY-REPORT-OBEYFILE**

  The fully-qualified file name of the obey file to run the Daily Transaction Volume Report Program. The format is xxDRmmdd, where 'xx' specifies the type of BASE24 product (AT, PS , TR, or RB for BASE24-atm, BASE24-pos, BASE24-teller, or BASE24-telebanking/billpay respectively).

  REQUIRED ASSIGN: Yes

- **DAILY-REPORT-PROGRAM**

  The fully-qualified file name of the object file to run the Daily Transaction Volume Report Program. The format is xxDRPT where 'xx' specifies the type of BASE24 product (AT, PS, TR or RB for BASE24-atm, BASE24-pos, BASE24-teller, or BASE24-telebanking/billpay respectively).

  REQUIRED ASSIGN: Yes

MONTHLY-REPORT-DESTINATION

The fully-qualified file name of the spooler location for the report produced by the Monthly Transaction Volume Report Program. The format is #xxMyymm, where 'xx' specifies the type of BASE24 product (AT, PS,TR, or RB for BASE24-atm, BASE24-pos, BASE24-teller or BASE24-tb respectively).

REQUIRED ASSIGN: Yes

MONTHLY-REPORT-OBEYFILE

The fully-qualified file name of the obey file to run the Monthly Transaction Volume Report Program. The format is xxMRyymm, where 'xx' specifies the type of BASE24 product (AT, PS, TR, or RB for BASE24-atm, BASE24-pos, BASE24-teller, or BASE-24-telebanking/billpay respectively).

REQUIRED ASSIGN: Yes

MONTHLY-REPORT-PROGRAM

The fully-qualified file name of the object file to run the Monthly Transaction Volume Report Program. The format is BAMRPT for BASE24-atm, BASE24-pos and BASE24-teller. The format is RBMRPTLO for BASE24-telebanking/billpay.

REQUIRED ASSIGN: Yes

MTAF and/or RBAF

The fully-qualified file name of the Monthly Totals Accumulation File. The format is xxMTyymm, where 'xx' specifies the type of BASE24 product (AT, PS, TR, or RB for BASE24-atm, BASE24-pos, BASE24-teller, or BASE24-tb respectively). The name of the subvolume where this file resides must be 8 characters long. The MTAF/RBAF requires an accompanying template file. Remote banking uses theTB-RBAF assign while all other products use XXX-MTAF (where XXX is ATM, POS, or TLR.

REQUIRED ASSIGN: Yes

2.2.2 LCONF Params

The LGNT-CUTOVER param is already present in the LCONF and is necessary for the TBP analysis tool to run successfully.

- **POS-ELEC-FOLLOW-UP-OPT**

  POS-ELEC-FOLLOW-UP-OPT controls how Electronic Follow-up transactions received through the VISA Interchange Interface are tabulated. If this param exists, the PTLF will be examined. If the VISA switch token exists and the STIP field is “1901”, the transaction will be considered to be an Electronic Follow-up transaction. If this param does not exist, a draft capture flag value of “3” will be the only indicator of an Electronic Follow-up transaction.

  REQUIRED PARAM: No
  DEFAULT VALUE: Y

- **DATE-DISPLAY-TYPE**

  DATE-DISPLAY-TYPE is an existing param used by reports, device handlers and Pathway. It indicates format for month and day for a date on daily reports (1 = DDMM, 2 = MMDD). The valid values are 1 and 2.

  REQUIRED PARAM: No
  DEFAULT VALUE: 2

- **MULT-LN-ACQ-TRAN-COUNTER**

  Indicates that transactions routed between two logical networks using CPF SPROUTE routing should be included in the count for the reports. “Y”, indicates to count the corresponding transactions. “N”, indicates to not count the corresponding transactions.

  REQUIRED PARAM: No
  DEFAULT VALUE: Y

One copy of the following LCONF params for each BASE24 product is at the customer site that is being monitored. **The param names are preceded by the product name; for example, ATM-, POS- , TLR-, or TB-.**

- **DLY-RPT-STARTUP**

  The time of day that the Daily and Monthly Transaction Volume Report Programs will run. The valid values are in the range 00:00 - 23:59.

  REQUIRED PARAM: Yes
  DEFAULT VALUE : None

- **RPT-CPU**

  The CPU in which the reports are run. The valid values are in the range 0 - n-1, where 'n' specifies the number of CPUs in the system.

  REQUIRED PARAM: No
  DEFAULT VALUE : The CPU of the RIP

- **RPT-PRI**

  The priority at which the reports are run. The valid values are in the range 1 -199.

  REQUIRED PARAM: No
  DEFAULT VALUE : The Priority of the RIP

2.3 Network Environment Source File (NEFS)

This section explains how the BASE24 processes, stations, lines and devices are defined for the TBP report programs. Only the minimum declaration requirements are described.

2.3.1 Process Declaration

The following process declaration is for the Report Initiator Process (RIP). The DEFINES element specifies whether TACL DEFINE values are passed from the XPNET process to the application process. It is applicable only for XPNET 2.0 or greater systems. The NSKSTARTSEQ element is for XPNET 3.0 systems only.

```
PROCESS   P1A^RIP:
          OBJECT                  "$volume.subvolume.RIP",
          LIBRARY,
          BOJ,
          DEFINES,
          NSKSTARTSEQ             OFF,
          PRIORITY                100,
          PPD NAME IS             "$PIRP",
          CPU                     0,
          LOGICAL NETWORK IS      PRO1:
```

2.4 Running the TBP Analysis Suite

Perform the following steps to run the TBP analysis tool:

1. Add the assigns and params to the LCONF.

2. Add the RIP process declaration to the network. A spare process may be used if available. In an XPNET 2.1 or earlier environment, if no spare process exists, the network must be stopped and the RIP process declaration must be added to the NEFS. In an XPNET 3.0 environment, a new process may be configured without stopping the network.

3. Create a template for the Monthly Totals Accumulation File (MTAF) and or Remote Banking Accumulation File (RBAF). To do so, volume to the template subvolume and use the MTTPLT file as an input file to FUP to create the template, XXMTYYMM and/or RBMTYYMM.

4. Bring up the network. The RIP process will start automatically as it has been configured as a Beginning Of Job (BOJ) process. If a spare process was used, start the process from the NCS screen. If the RIP process initialized successfully, the following informational message is displayed.

INITIALIZATION COMPLETE FOR RIP PROCESS

Otherwise, an error message is logged and the process is abended.

On startup, the RIP process creates a daily report obey file and an MTAF and/or RBAF for the last completed business date. If the last completed business date is the first of the month, the RIP process also creates a monthly report obey file for the previous month. The RIP process delays the processing until either the logical network cutover time or daily report startup time is reached.

At the logical network cutover time, the RIP process creates a daily report obey file for the business date that has just been cutover. If the date is the first of the month, a monthly report obey file is created for the previous month. If the date that has been cutover to is the first of the month, an MTAF and/or RBAF is created for that month.

At daily report startup time, a TACL is started to run the daily report obey file that was created at the logical network cutover. If the current business day is the second of the month, then the daily report for the first of the month is run. This means that all the daily reports for the previous month have been run and, therefore, the monthly report for that month can be run. In this case, a TACL is started to run the monthly report obey file that was created at logical network cutover on the first of the month.

The Transaction Based Pricing Report Program creates a report and sends it to the spooler location specified in the LCONF MONTHLY-REPORT-DESTINATION assign.

2.5 Acceleration Steps

Customers may wish to accelerate the TBP objects for better performance. Listed below are steps required to accelerate the TBP objects.

1. Back up the TBP objects.

2. Volume to the TBP subvolume.

3. Create an edit file, TBPAXCEL.

4. Add the lines similar to the following for each object to be accelerated.

```
AXCEL/OUT $S.#ATDRPT, CPU 0, PRI 100/ATDRPT, ATDRPT:&
   USERLIB $SYSACI.SPANNET.HSKELB
```

5. Exit the editor.

6. Obey TBPAXCEL.

7. Accelerated objects will be created having the same names as the unaccelerated objects.

3 MTAF

The Monthly Totals Accumulation File (MTAF) is a key-sequenced file with the date field as the key. The date format is yymmdd. The format of the file name is:

xxMTyymm

where 'xx' specifies the type of BASE24 product; for example, AT, PS or TR specifies BASE24-atm, BASE24-pos or BASE24-teller respectively.

'yymm' specifies the year and month that the MTAF is for.

One MTAF is created for each product by the RIP process when it is first started and on the first day of each month. If an MTAF exists on the first day of the month, it will be purged and a new MTAF will be created. The daily report programs either add or update daily records in their respective MTAFs.

4 RBAF

The Remote Banking Accumulation (RBAF) is a key-sequenced file with the date field as the key. The date format is yymmdd. The format of the file name is:

RBMTyymm

where 'RB' specifies Remote Banking.

'yymm' specifies the year and month that the RBAF is for.

One RBAF is created for Telebanking and Billpay, not one for each product, by the RIP process when it is first started and on the first day of each month. If an RBAF exists on the first day of the month, it will be purged and a new RBAF will be created. The daily report program either adds or updates daily records to the RBAF.

5 Reports

Two types of reports are produced by the TBP programs:

- **Daily Transaction Volume Report**

  The Daily Transaction Volume Report program produces this report from the daily analysis of the Transaction Log File (TLF, PTLF, TTLF or ITLF). A report of the totals may be printed, if required.

- **Monthly Transaction Volume Report**

  The Monthly Transaction Volume Report program produces this report from the monthly analysis of the Monthly Totals Accumulation File (MTAF) and/or the Remote Banking Accumulation File (RBAF). The MTAF and/or the RBAF contains one record for each day of the month. A report of the totals may be printed, if required

  Each record contains counts of customer and non-customer transactions from daily transaction log files and the daily report version number. Customer transactions are divided into categories by source and authorizer and whether they were approved or denied. If a record contains a version number different from that of the monthly program, a message will be displayed and written to the report.

  Only authorized network operators can access the MTAF and/or RBAF or initiate programs that access the MTAF and/or RBAF.

5.1 Sample Report Obey Files

Following are examples of the daily and monthly report obey files:

```
run \CFS.$DATA13.BA60TBP.ATDRPT         &
     /in \CFS.$DATA13.R600CNTL.L1CONF,  &
     out \CFS.$S.#ATD0430,              &
     cpu 0 , pri 101/DATE 010430

run \CFS.$DATA13.BA60TBP.BAMRPT         &
     /in \CFS.$DATA13.R600CNTL.L1CONF,  &
     out \CFS.$S.#PSM0106,              &
     cup 0, pri 125/FILE PSMT0106
```

5.2 Examples of Transaction Volume Reports

The following pages contain layouts of the ATM/POS/RB/TLR Daily/Monthly Transaction Volume Reports.

The report example below is the ATM Daily Transaction Volume Report.

Source Determined by Originator Code

Destination Determined by Responder Code

```
Page   1  S      K9     A T M   D A I L Y   T R A N S A C T I O N   V     O M E   R E P O R T - 09 Logical network TES7  10/06/05

SOURCE                                                  DESTINATION
------                                                  -----------

                Base24 & Force Post                     Host                    Interchange                    Total
                -------------------                     ----                    -----------
                Approved   Denied    Total   Approved   Denied    Total   Approved   Denied    Total
                --------   ------    -----   --------   ------    -----   --------   ------    -----

Base24             0         0         0        0         0         0        0         0         0            0
Host               0         0         0        0         0         0        0         0         0            0
Intchg             0         0         0        0         0         0        0         0         0            0
                --------   ------    -----   --------   ------    -----   --------   ------    -----   ---------
Total              0         0         0        0         0         0        0         0         0            0

                Secondary Service                                                                          Total
                -----------------                                                                          -----
                Approved   Denied    Total
                --------   ------    -----

Base24             0         0         0                                                                       0
Host               0         0         0                                                                       0
Intchg             0         0         0                                                                       0
                --------   ------    -----                                                              ---------
Total              0         0         0                                                                       0
                ========   ======    =====                                                              =========

Invalid-responder   Invalid-originator                                                                    Total
-----------------   -------------------                                                                    -----
        0                   0                                                                                 0
OTHER RECORD CATEGORIES

Reversals        Pre-auth-comps      Admin-trans       Exceptions       Non-transactions                  Total
---------        --------------      -----------       ----------       ----------------                  -----
    0                  0                  0                 0                  0                              0
                                                                                                   Total TLF records
                                                                                                   -----------------
                                                                                                          0

TOTAL TLF RECORDS                                                    0
TOTAL OTHER RECORD CATEGORIES                                        0
                                                             ----------
TOTAL TRANSACTIONS PROCESSED                                         0
                                                             ==========
```

A

AA

B

C

I

D

E

F

G

H

J

K

K

J

K - J

Everything on the Log File

Minus Excluded Records

Equals TBP Transaction Count

A	General financial transactions
AA	Secondary Service transactions (logged to TLF by the BASE24-atm Transaction Context Manager (ATCM) process
B	Transactions that have an unrecognized or invalid authorizer (invalid responder code)
C	Transactions that have an unrecognized or invalid source (invalid originator code)
D	Reversals (message type 0420)
E	Pre-auth Completions (n/a for ATM transactions)
F	Admin trans (e.g., ATM balancing, settlement). Record type = 04
G	All records with Record Type = 20, 21, 22 ("invalid data")
H	TLF "Record Zero", an Extract processing control record, plus anything else that cannot be categorized
I	Row Total of records with invalid responder/originator
J	Row Total of "Other Record Categories" (i.e., to be excluded)
K	Column Total, A+I+J. Equals total number of all TLF Records

The report example below is the ATM Monthly Transaction Volume Report.

Page 1 System \K9 A T M M O N T H L Y T R A N S A C T I O N V O L U M E R E P O R T - 09 Logical network TES7 08/05

SOURCE / DESTINATION

	Base24 & Force Post			Host			Interchange			Total
	Approved	Denied	Total	Approved	Denied	Total	Approved	Denied	Total	
Base24	0	0	0	0	0	0	0	0	0	0
Host	0	0	0	0	0	0	0	0	0	0
Intchg	0	0	0	0	0	0	0	0	0	0
Total	0	0	0	0	0	0	0	0	0	0

	Secondary Service			Total
	Approved	Denied	Total	
Base24	0	0	0	0
Host	0	0	0	0
Intchg	0	0	0	0
Total	0	0	0	0

Invalid-responder	Invalid-originator	Total
0	0	0

E

OTHER RECORD CATEGORIES

Reversals	Pre-auth-comps	Admin-trans	Exceptions	Non-transactions
0	0	190	0	1

Batch Upload	Elect Followup	Total
0	0	191

L M

Total TLF records
191

TOTAL TLF RECORDS 191
TOTAL OTHER RECORD CATEGORIES 191

TOTAL TRANSACTIONS PROCESSED 0

E	Pre-Auth Completions; (n/a for ATM transactions)
L	Batch Upload (n/a for ATM transactions)
M	Electronic Followup: (n/a for ATM transactions)

The report example below is the POS <Daily/Monthly> Transaction Volume Report.

```
Page   1  System \K9      P O S  M O N T H L Y   T R A N S A C T I O N   V O L U M E   R E P O R T - 09  Logical network TES7  09/05

SOURCE                                                  DESTINATION
------                                                  -----------

            Base24 & Force Post                          Host                          Interchange                      Total
          ---------------------------------  ---------------------------------  ---------------------------------  -----------
           Approved     Denied       Total     Approved     Denied       Total     Approved     Denied       Total
          ----------- ----------- -----------  ----------- ----------- -----------  ----------- ----------- -----------

Base24          0           0           0           0           0           0           0           0           0           0
Host            0           0           0           0           0           0           0           0           0           0
Intchg          0           0           0           0           0           0           0           0           0           0
          ----------- ----------- -----------  ----------- ----------- -----------  ----------- ----------- -----------  -----------
Total           0           0           0           0           0           0           0           0           0           0

            Secondary Service                                                                                              Total
          ---------------------------------                                                                            -----------
           Approved     Denied       Total
          ----------- ----------- -----------

Base24          0           0           0                                                                                       0
Host            0           0           0                                                                                       0
Intchg          0           0           0                                                                                       0
          ----------- ----------- -----------                                                                            -----------
Total           0           0           0                                                                                       0
          =========== =========== ===========                                                                            ===========

      Invalid-responder   Invalid-originator                                                                                Total
      -----------------   -----------------                                                                            -----------
                  0                   0                                                                                       0

OTHER RECORD CATEGORIES

         Reversals          Pre-auth-comps      Admin-trans         Exceptions       Non-transactions
       ---------------     ---------------    ---------------    ---------------    ---------------
                  0                   0                  0                  0                  0

       Batch Upload        Elect Followup                                                                                   Total
       ----------------    ----------------                                                                             -----------
                  0                   0                                                                                       0
                                                                                                                   Total PTLF records
                                                                                                                   ------------------
                                                                                                                                   0

                                                  TOTAL PTLF RECORDS                          0
                                                  TOTAL OTHER RECORD CATEGORIES               0
                                                                                      -----------
                                                  TOTAL TRANSACTIONS PROCESSED                0
                                                                                      ===========
```

E	**Pre-Auth Completions; Tran Code = 12**
L	**Batch Upload:Hypercom Device Handler; entire batch uploads when out of balance (i.e., retransmission). Draft Capture flag = 3.**
M	**Electronic Followup: VISA2 Device Handler; follow-up component of two-step auth and capture transaction.**

The report example below is the RB <Daily/Monthly> Transaction Volume Report.

```
Page   1  System \R       R B     D A I L Y   T R A N S A C T I O N   V O L U M E   R E P O R T   09 Logical network TES4 6/12/98

 SOURCE                                                  DESTINATION
 ------                                                  -----------
                                 BASE24-telebanking                          BASE24-billpay
                          ----------------------------------          ----------------------------------
                           Approved     Denied       Total             Approved     Denied       Total
                          ----------  ----------  ----------          ----------  ----------  ----------
 CSR                               0           0           0                   0           0           0
 EBMS                              0           0           0                   0           0           0
 Host                              0           0           0                   0           0           0
 Other                             0           0           0                   0           0           0
 RBSI                              0           0           0                   0           0           0
 STI                               0           0           0                   0           0           0
 XPBP                              0           0           0                   0           0           0
                          ----------  ----------  ----------          ----------  ----------  ----------
 Total                             0           0           0                   0           0           0
                          ==========  ==========  ==========          ==========  ==========  ==========

                                   Host                                   Interchange                                  Total
                          ----------------------------------          ----------------------------------          ----------
                           Approved     Denied       Total             Approved     Denied       Total
                          ----------  ----------  ----------          ----------  ----------  ----------
 CSR                               0           0           0                   0           0           0                   0
 EBMS                              0           0           0                   0           0           0                   0
 Host                              0           0           0                   0           0           0                   0
 Other                             0           0           0                   0           0           0                   0
 RBSI                              0           0           0                   0           0           0                   0
 STI                               0           0           0                   0           0           0                   0
 XPBP                              0           0           0                   0           0           0                   0
                          ----------  ----------  ----------          ----------  ----------  ----------          ---------
 Total                             0           0           0                   0           0           0                   0
                          ==========  ==========  ==========          ==========  ==========  ==========          ==========

 OTHER RECORD CATEGORIES
         Invalid-responder   Invalid-originator                                                                        Total
         -----------------   ------------------                                                                     ----------
                    97                   0                                                                                   0

             Reversals      Pre-auth-comps      Admin-trans       Exceptions     Non-transactions  Immediate Billpay        Total
         ----------------  ----------------  ----------------  ----------------  ----------------  -----------------   ----------
                   0                 0                 0                 0                 0                   0                 0

                                                                                                                   Total ITLF
                                                                                 records
                                                                                                                   ----------
                                                                                                                            0
                                          TOTAL ITLF RECORDS                          0
                                          TOTAL OTHER RECORD CATEGORIES               0
                                                                             -----------
                                          TOTAL TRANSACTIONS PROCESSED                0
```

The report example below is the TLR Daily Transaction Volume Report.

```
Page   1  System \R      T L R    D A I L Y   T R A N S A C T I O N   V O L U M E   R E P O R T - 09 Logical network R514 7/02/98

 SOURCE                                                DESTINATION
 ------                                                -----------

                 Base24 & Force Post                        Host                          Interchange                    Total
        ----------------------------------  ----------------------------------  ----------------------------------  ----------
           Approved     Denied       Total     Approved     Denied       Total     Approved     Denied       Total
        ----------- ----------- -----------  ----------- ----------- -----------  ----------- ----------- -----------
 Base24          0           0           0            0           0           0            0           0           0           0
 Host            0           0           0            0           0           0            0           0           0           0
 Intchg          0           0           0            0           0           0            0           0           0           0
        ----------- ----------- -----------  ----------- ----------- -----------  ----------- ----------- -----------  ----------
 Total           0           0           0            0           0           0            0           0           0           0
        =========== =========== ===========  =========== =========== ===========  =========== =========== ===========  ==========

       Invalid-responder   Invalid-originator                                                                              Total
       -----------------   ------------------                                                                            ----------
                   0                   0                                                                                          0

 OTHER RECORD CATEGORIES
          Reversals         Pre-auth-comps        Admin-trans         Exceptions       Non-transactions                   Total
        ----------------    ----------------    ----------------    ----------------    ----------------                ----------
                   0                   0                   0                   0                   0                               0

                                                                                                                        Total TTLF
                                                                  records
                                                                                                                        -----------------
                                                                                                                                       0


                                      TOTAL TTLF RECORDS                          0
                                      TOTAL OTHER RECORD CATEGORIES               0
                                                                            -----------
                                      TOTAL TRANSACTIONS PROCESSED                0
```

The report example below is the TLR Monthly Transaction Volume Report.

```
Page   1  System \K9      T L R  M O N T H L Y   T R A N S A C T I O N   V O L U M E   R E P O R T - 09  Logical network TES7  10/05

 SOURCE                                                  DESTINATION
 ------                                                  -----------
                 Base24 & Force Post                        Host                          Interchange                     Total
          ----------------------------------  ----------------------------------  ----------------------------------  -----------
           Approved     Denied       Total     Approved     Denied       Total     Approved     Denied       Total
          -----------  -----------  -----------  -----------  -----------  -----------  -----------  -----------  -----------

 Base24           0            0            0            0            0            0            0            0            0            0
 Host             0            0            0            0            0            0            0            0            0            0
 Intchg           0            0            0            0            0            0            0            0            0            0
          -----------  -----------  -----------  -----------  -----------  -----------  -----------  -----------  -----------  -----------
 Total            0            0            0            0            0            0            0            0            0            0

                Secondary Service                                                                                          Total
          ----------------------------------                                                                           -----------
           Approved     Denied       Total
          -----------  -----------  -----------

 Base24           0            0            0                                                                                   0
 Host             0            0            0                                                                                   0
 Intchg           0            0            0                                                                                   0
          -----------  -----------  -----------                                                                        -----------
 Total            0            0            0                                                                                   0
          ===========  ===========  ===========                                                                        ===========

       Invalid-responder   Invalid-originator                                                                                Total
       -----------------   ------------------                                                                           -----------
                   0                   0                                                                                           0

 OTHER RECORD CATEGORIES

           Reversals          Pre-auth-comps      Admin-trans          Exceptions       Non-transactions
        ----------------     ----------------   ----------------    ----------------    ----------------
                   0                   0                   0                   0                   1

        Batch Upload        Elect Followup                                                                                   Total
       -----------------   ------------------                                                                           -----------
                   0                   0                                                                                           1
                                                                                                                   Total TTLF records
                                                                                                                   ------------------
                                                                                                                                   1

                                   TOTAL TTLF RECORDS                          1
                                   TOTAL OTHER RECORD CATEGORIES               1
                                                                       -----------
                                   TOTAL TRANSACTIONS PROCESSED                0
                                                                      =============
```

AA	**Secondary Service transactions (n/a for TLR transactions)**
E	**Pre-Auth Completions; (n/a for TLR transactions)**
L	**Batch Upload (n/a for TLR transactions)**
M	**Electronic Followup: (n/a for TLR transactions)**

APPENDIX A - LOG MESSAGE FORMAT

Log messages, unless otherwise stated, are written to the requester screens or the BASE24 Disk Log File and, optionally, on other logging devices as defined by the BASE24 log configuration.

Log messages are generated to report an error condition or information. Error messages may require corrective action before BASE24 can function properly.

The standard format for BASE24 log messages is:

YY/MM/DD HH:MM:SS (X) <PROCESS ID> ~OBJECT> MMMM RRRR <TEXT>

WHERE: 'YY/MM/DD' specifies the date the message was sent.
'HH:MM:SS' specifies the time the message was sent.
'(X)' specifies the severity code where: (0) or (I) Status (Informational)
(1) or (W) Warning
(2) or (E) Error
(3) or (C) Critical

'<PROCESS ID>' specifies the symbolic process name.
'<OBJECT>' specifies a 1 byte string defined and passed by the process.
'MMMM' specifies the message number.
'RRRR' specifies the routing code.
'<TEXT>' specifies the text of the message.

For example:

92/01/02 12:00:00 (O) RIP 0113 0002 INITIALIZATION COMPLETE FOR RIP PROCESS

To avoid the repetition of showing the date, time, process identifier, object and routing code for each message; only the message number, severity code and the text of the message are documented. Therefore, the display format of log messages adopted for documentation is:

MMMM (X) <TEXT>

For example:

0113	(O)	INITIALIZATION COMPLETE FOR RIP PROCESS

For each log message, additional information is given:

- the cause or reason for displaying the message.
- the remedy or solution for the message, if relevant.

Also, the following support/service personnel may need to be contacted:

- Operations Personnel

  In-house operations staff who are responsible for initiating fault action for a particular problem.

- HELP24

  ACI personnel who are responsible for solving customer queries and problems.

APPENDIX B - RIP PROCESS MESSAGES

0001	(3)	Invalid LN-CUTOVER specified in lconf \f

WHERE: '\f' specifies the LCONF name.
CAUSE: The LN-CUTOVER param in the indicated LCONF does not contain valid data.
REMEDY: Correct the LN-CUTOVER param in the LCONF and restart the process.

0039	(3)	Open of lconf file \f failed - error \e

WHERE: '\f' specifies the LCONF name.
'\e' specifies the error number.
CAUSE: An error occurred during the open of the indicated LCONF.
REMEDY: Correct the error and restart the process.

0040	(2)	Warmboot of process \s is now completed

WHERE: '\s' specifies the process name.
CAUSE: The warmboot of the indicated process has completed successfully.
REMEDY: No action is required.

0044	(2)	TACL \f, open error \e

WHERE: '\f' specifies the TACL process name.
'\e' specifies the error number.
CAUSE: An error occurred while trying to open the TACL process to run a report obey file.
REMEDY: Run the report manually.

0045	(2)	TACL \f, write error \e

WHERE: '\f' specifies the TACL process name.
'\e' specifies the error number.
CAUSE: An error occurred while trying to write to the TACL process to run a report obey file.
REMEDY: Run the report manually.

0046	(2)	TACL program \p, newprocess error \e

WHERE: '\p' specifies the TACL process name.
'\e' specifies the error number.
CAUSE: An error occurred while trying to create the TACL process to run a report obey file.
REMEDY: Run the report manually.

0047	(1)	TACL program \f may not have run the TBP reports.

WHERE: '\f' specifies the TACL process name.
CAUSE: An error occurred while the TACL process was trying to run a report obey file.
REMEDY: Run the report manually.

0048	(1)	Possible error in config file assign for \f.

WHERE: '\f' specifies the LCONF name.
CAUSE: An invalid LCONF assign or assigns were encountered when running the reports.
REMEDY: Correct the values of the LCONF assigns(s).

0049	(0)	TACL \f started to run the report programs

WHERE: '\f' specifies the TACL process name.
CAUSE: The indicated TACL process has been started to run the TBP report program.
REMEDY: No action is required.

0055	(0)	Reports can be run by using the \f obeyfile

WHERE: '/f' specifies the obey file name
CAUSE: The indicated obey file can be used to manually rerun the report.
REMEDY: No action is required.

0061	(2)	TACL will not be started to run the ATM daily report program

CAUSE: An error occurred while creating the ATM daily report obey file, so the report will not be run automatically.
REMEDY: Rectify the situation and warmboot the RIP process or run the report program manually.

0062	(2)	TACL will not be started to run the ATM monthly report program

CAUSE: An error occurred while creating the ATM monthly report obey file, so the report will not be run automatically.
REMEDY: Rectify the situation and warmboot the RIP process or run the report program manually.

0063	(2)	TACL will not be started to run the POS daily report program

CAUSE: An error occurred while creating the POS daily report obey file, so the report will not be run automatically.
REMEDY: Rectify the situation and warmboot the RIP process or run the report program manually.

0064	(2)	TACL will not be started to run the POS monthly report program

CAUSE: An error occurred while creating the POS monthly report obey file, so the report will not be run automatically.
REMEDY: Rectify the situation and warmboot the RIP process or run the report program manually.

0065	(2)	TACL will not be started to run the TLR daily report program

CAUSE: An error occurred while creating the Teller daily report obey file, so the report will not be run automatically.
REMEDY: Rectify the situation and warmboot the RIP process or run the report program manually.

0066	(2)	TACL will not be started to run the TLR monthly report program

CAUSE: An error occurred while creating the Teller monthly report obey file, so the report will not be run automatically.
REMEDY: Rectify the situation and warmboot the RIP process or run the report program manually.

0071	(2)	ATM daily report program's TACL obey file \f purge error \e

WHERE: '\f' specifies the obey file name.
'\e' specifies the error number.
CAUSE: An error occurred while trying to purge the ATM daily report obey file, so the report will not be run automatically.
REMEDY: Rectify the situation and warmboot the RIP process or run the report program manually.

0072	(2)	ATM monthly report program's TACL obey file \f purge error \e

WHERE: '\f' specifies the obey file name.
'\e' specifies the error number.
CAUSE: An error occurred while trying to purge the ATM monthly report obey file, so the report will not be run automatically.
REMEDY: Rectify the situation and warmboot the RIP process or run the report program manually.

0073	(2)	POS daily report program's TACL obey file \f purge error \e

WHERE: '\f' specifies the obey file name.
'\e' specifies the error number.
CAUSE: An error occurred while trying to purge the POS daily report obey file, so the report will not be run automatically.
REMEDY: Rectify the situation and warmboot the RIP process or run the report program manually.

0074	(2)	POS monthly report program's TACL obey file \f purge error \e

WHERE: '\f' specifies the obey file name.
'\e' specifies the error number.
CAUSE: An error occurred while trying to purge the POS monthly report obey file, so the report will not be run automatically.
REMEDY: Rectify the situation and warmboot the RIP process or run the report program manually.

0075	(2)	TLR daily report program's TACL obey file \f purge error \e

WHERE: '\f' specifies the obey file name.
'\e' specifies the error number.
CAUSE: An error occurred while trying to purge the Teller daily report obey file, so the report will not be run automatically.
REMEDY: Rectify the situation and warmboot the RIP process or run the report program manually.

0076	(2)	TLR monthly report program's TACL obey file \f purge error \e

WHERE: '\f' specifies the obey file name.
'\e' specifies the error number.
CAUSE: An error occurred while trying to purge the Teller monthly report obey file, so the report will not be run automatically.
REMEDY: Rectify the situation and warmboot the RIP process or run the report program manually.

0081	(2)	ATM MTAF \f purge error \e

WHERE: '\f' specifies the ATM MTAF file name.
'\e' specifies the error number.
CAUSE: An error occurred while trying to purge an old ATM Monthly Transaction Accumulation File on the first day of the month.
REMEDY: Rectify the situation and warmboot the RIP process.

0082	(2)	POS MTAF \f purge error \e

WHERE: '\f' specifies the POS MTAF file name.
'\e' specifies the error number.
CAUSE: An error occurred while trying to purge an old POS Monthly Transaction Accumulation File on the first day of the month.
REMEDY: Rectify the situation and warmboot the RIP process.

0083	(2)	TLR MTAF \f purge error \e

WHERE: '\f' specifies the Teller Monthly Transaction accumulation file name.
'\e' specifies the error number.
CAUSE: An error occurred while trying to purge an old Teller Monthly Accumulation File on the first day of the month.
REMEDY: Rectify the situation and warmboot the RIP process.

0084	(2)	RB RBAF \f purge error \e

WHERE: '\f' specifies the Remote Banking RBAF file name.
'\e' specifies the error number.
CAUSE: An error occurred while trying to purge an old Remote Banking Accumulation File on the first day of the month.
REMEDY: Rectify the situation and warmboot the RIP process.

0090	(3)	\? error \e for entry **********

WHERE: '\?' specifies the error text.
'\e' specifies the error number.
'**********' specifies the failed function.
CAUSE: An error occurred on the indicated LCONF entry.
REMEDY: Rectify the situation, if required.

0091	(3)	\? error # for entry **********

WHERE: '\?' specifies the error text.
'#' specifies the error number.
'**********' specifies the failed function.
CAUSE: An error occurred on the indicated LCONF entry.
REMEDY: Rectify the situation, if required.

0093	(3)	Invalid template file name in lconf \f for entry *

WHERE: '\f' specifies the LCONF name.
'*' specifies the failed function.
CAUSE: An error occurred in the template file for the indicated LCONF entry.
REMEDY: Correct the LCONF entry.

0094	(1)	WARNING - \? is missing from the lconf \f.

WHERE: '\?' specifies the error text.
'\f' specifies the LCONF name.
CAUSE: An optional assign or param is missing from the indicated LCONF.
REMEDY: Add the missing assign or param, if required, and warmboot the process.

0095	(3)	\? is missing from the lconf \f, RIP stopped

WHERE: '\?' specifies the error text.
'\f' specifies the LCONF name.
CAUSE: A required assign or param is missing from the indicated LCONF so the RIP process has stopped.
REMEDY: Add the missing assign or param, if required, and warmboot the process.

0096	(1)	Invalid \? specified in LCONF \f, defaulting to #

WHERE: '\?' specifies the error text.
'\f' specifies the LCONF name.
'#' specifies the default value.
CAUSE: An invalid entry is processed in the indicated LCONF so the default value is used.
REMEDY: If the default value is not acceptable, correct the LCONF entry and warmboot the RIP process.

0097	(3)	There are no MTAF or RBAF files in the LCONF, RIP stopped.

CAUSE: There are no assigns for the MTAF or RBAF.
REMEDY: Add an assign for the ATM, POS or TLR MTAF or for the RBAF and restart the process.

0099	(3)	Invalid filename in lconf \f for **********

WHERE: '\f' specifies the LCONF name.
'**********' specifies the filename.
CAUSE: An error occurred in the indicated LCONF entry.
REMEDY: Correct the LCONF entry.

0100	(3)	Invalid report startup time specified in \f

WHERE: '\f' specifies the LCONF name.
CAUSE: An error occurred in the report startup time in the indicated LCONF entry.
REMEDY: Correct the LCONF entry.

0101	(2)	Unable to create, using the template file, error \e

WHERE: '\e' specifies the error number.
CAUSE: An error occurred while the RIP process was accessing the template file to create the MTAF and/or RBAF.
REMEDY: Rectify the situation with the MTAF and/or RBAF template file and warmboot the RIP process.

0112	(3)	Error accessing \f, RIP process cannot continue

WHERE: '\f' specifies the file name.
CAUSE: An error occurred during the access of the indicated file.
REMEDY: Rectify the situation.

0113	(0)	INITIALIZATION COMPLETE FOR RIP PROCESS

CAUSE: The RIP process has initialized successfully.
REMEDY: No action is required.

0117	(2)	Msg received from %@@ with unknown type \\\\: dropped

WHERE: '%@@' specifies the process name.
'\\\\' specifies the message type.
CAUSE: The RIP process has received an unknown message type from the specified process. The message has been dropped.
REMEDY: No action is required.

0118	(2)	Invalid message received from \f

WHERE: '\f' specifies the file name.
CAUSE: An invalid message was received from the indicated process.
REMEDY: No action is required.

0119	(2)	Interprocess communication failure, File \f, Error \e

WHERE: '\f' specifies the file name.
'\e' specifies the error number.
CAUSE: An error occurred when a process tried to communicate with another process.
REMEDY: No action is required.

0125	(2)	Unable to create, using the template file, reason \?

WHERE: '\?' specifies the error text.
CAUSE: An error occurred while the RIP process was accessing the template file to create the MTAF and/or RBAF.
REMEDY: Rectify the situation with the MTAF and/or RBAF template file and warmboot the RIP process.

0183	(2)	Abnormal deletion of process \f

WHERE: '\f' specifies the file name.
CAUSE: An error occurred when a process tried to communicate with another process.
REMEDY: No action is required.

APPENDIX C - DAILY REPORT PROGRAM MESSAGES

Date parameter must be specified; PROGRAM ABORTED

CAUSE: The DATE parameter was not passed to the report program in the startup message.
REMEDY: Ensure that the DATE parameter is passed correctly and then rerun the report program manually.

ERROR (#) opening LCONF (\f)

WHERE: '#' specifies the error number.
'\f' specifies the LCONF name.
CAUSE: An error occurred when the process tried to open the indicated LCONF.
REMEDY: Resolve the error in the LCONF and manually rerun the report program.

ERROR (\e) reading record in TLF; REPORT ABORTED

WHERE: '\e' specifies the error number.
CAUSE: An error occurred when the process tried to read the indicated TLF.
REMEDY: Resolve the error in the TLF and manually rerun the report program.

ERROR \e opening \f for output, PROGRAM ABORTED

WHERE: '\e' specifies the error number.
'\f' specifies the file name.
CAUSE: An error occurred when the process tried to open the indicated file.
REMEDY: Resolve the error in the indicated file and manually rerun the report program.

Error writing line # to page # of report

WHERE: '#' specifies the error line number.
'#' specifies the error page number.
CAUSE: An error occurred when the process tried to write the report output to the spooler.
REMEDY: Resolve the error in the spooler file and manually rerun the report program.

FATAL ERROR (\e) opening MTAF (\f); PROGRAM ABORTED

WHERE: '\e' specifies the error number.
'\f' specifies the MTAF name.
CAUSE: An error occurred when the process tried to open the indicated MTAF.
REMEDY: Resolve the error in the MTAF and manually rerun the report program.

FATAL ERROR (\e) opening RBAF (\f); PROGRAM ABORTED

WHERE: '\e' specifies the error number.
'\f' specifies the RBAF name.
CAUSE: An error occurred when the process tried to open the indicated RBAF.
REMEDY: Resolve the error in the RBAF and manually rerun the report program.

FATAL ERROR (\e) opening TLF (\f); PROGRAM ABORTED

WHERE: '\e' specifies the error number.
'\f' specifies the TLF name.
CAUSE: An error occurred when the process tried to open the indicated TLF.
REMEDY: Resolve the error in the TLF and manually rerun the report program.

FATAL ERROR (\e) positioning on MTAF (\f); PROGRAM ABORTED

WHERE: '\e' specifies the error number.
'\f' specifies the MTAF name.
CAUSE: An error occurred when the process tried to position in the indicated MTAF.
REMEDY: Resolve the error in the MTAF and manually rerun the report program.

FATAL ERROR (\e) reading from MTAF (\f); PROGRAM ABORTED

WHERE: '\e' specifies the error number.
'\f' specifies the MTAF name.
CAUSE: An error occurred when the process tried to read the indicated MTAF.
REMEDY: Resolve the error in the MTAF and manually rerun the report program.

FATAL ERROR (\e) reading from RBAF (\f); PROGRAM ABORTED

WHERE: '\e' specifies the error number.
'\f' specifies the RBAF name.
CAUSE: An error occurred when the process tried to read the indicated RBAF.
REMEDY: Resolve the error in the RBAF and manually rerun the report program.

FATAL ERROR (\e) writing to MTAF (\f); PROGRAM ABORTED

WHERE: '\e' specifies the error number.
'\f' specifies the MTAF name.
CAUSE: An error occurred when the process tried to write to the indicated MTAF.
REMEDY: Resolve the error in the MTAF and manually rerun the report program.

FATAL ERROR (\e) writing to RBAF (\f); PROGRAM ABORTED

WHERE: '\e' specifies the error number.
'\f' specifies the RBAF name.
CAUSE: An error occurred when the process tried to write to the indicated RBAF.
REMEDY: Resolve the error in the RBAF and manually rerun the report program.

Invalid date param value: **********; PROGRAM ABORTED

WHERE: '**********' specifies the invalid value of the DATE param.
CAUSE: An invalid DATE param was passed to the report program in the startup message.
REMEDY: Check the RIP process to ensure that the DATE param is passed correctly and then rerun the report program manually.

Invalid parameter: **********; PROGRAM ABORTED

WHERE: '**********' specifies the invalid value of the param.
CAUSE: The invalid param was passed to the report program in the startup message.
REMEDY: Check the RIP process to ensure that the startup message is correct and then rerun the report program manually.

TLF file empty; REPORT ABORTED

CAUSE: The indicated TLF to be analyzed contains no records.
REMEDY: Check that the correct TLF was being analyzed and the LCONF assign for the TLF is correct.

WARNING Param LOGICAL-NET not found in LCONF

CAUSE: The LCONF LOGICAL-NET param is missing and, therefore, the Logical Network is not displayed correctly on the report output.
REMEDY: If it is required that the Logical Network is shown correctly on the report output, ensure that the LCONF LOGICAL-NET param is added to the LCONF. Rerun the report program manually.

APPENDIX D - MONTHLY REPORT PROGRAM MESSAGES

ERROR (#) Open LCONF (\f) Failed (#)

WHERE: '#' specifies the error number.
'\f' specifies the LCONF name.
'#' specifies the error.
CAUSE: An error occurred when the process tried to open the indicated LCONF.
REMEDY: Resolve the error in the LCONF and manually rerun the report program.

ERROR (\e) reading record in MTAF; REPORT ABORTED

WHERE: '\e' specifies the error number.
CAUSE: An error occurred when the process tried to read the indicated MTAF.
REMEDY: Resolve the error in the MTAF and manually rerun the report program.

ERROR (\e) reading record in RBAF(\f); REPORT ABORTED

WHERE: '\e' specifies the error number.
‘\f’ specifies the RBAF name.
CAUSE: An error occurred when the process tried to read the indicated RBAF.
REMEDY: Resolve the error in the RBAF and manually rerun the report program.

ERROR \e opening \f for output; PROGRAM ABORTED

WHERE: '\e' specifies the error number.
'\f' specifies the file name.
CAUSE: An error occurred when the process tried to open the indicated file.
REMEDY: Resolve the error in the indicated file and manually rerun the report program.

ERROR MTAF Assign not found in LCONF

CAUSE: An error occurred when the process tried to open the indicated LCONF. The LCONF MTAF assign is missing and the report program cannot be run.
REMEDY: Add the MTAF assign to the LCONF and manually rerun the report program.

ERROR TB-RBAF Assign not found in LCONF

CAUSE: An error occurred when the process tried to open the indicated LCONF. The LCONF TB-RBAF assign is missing and the report program cannot be run.
REMEDY: Add the TB-RBAF assign to the LCONF and manually rerun the report program.

ERROR Output file must be specified; PROGRAM ABORTED

CAUSE: An output file was not passed to the report program in the startup message. An output file is required to run the monthly report program.
REMEDY: Check the RIP processing to ensure that the output file is passed correctly and then rerun the report program manually.

Error writing line # to page # of report

WHERE: '#' specifies the error line number.
'#' specifies the error page number.
CAUSE: An error occurred when the process tried to write the report output to the spooler.
REMEDY: Resolve the error in the spooler file and manually rerun the report program.

FATAL ERROR (\e) opening \f for output; PROGRAM ABORTED

WHERE: '\e' specifies the error number.
'\f' specifies the file name.
CAUSE: An error occurred when the process tried to open the indicated file.
REMEDY: Resolve the error in the indicated file and manually rerun the report program.

FATAL ERROR (\e) opening MTAF (\f); PROGRAM ABORTED

WHERE: '\e' specifies the error number.
'\f' specifies the MTAF name.
CAUSE: An error occurred when the process tried to open the indicated MTAF.
REMEDY: Resolve the error in the MTAF and manually rerun the report program.

FATAL ERROR (\e) opening RBAF(\f); PROGRAM ABORTED

WHERE: '\e' specifies the error number.
'\f' specifies the RBAF name.
CAUSE: An error occurred when the process tried to open the indicated RBAF.
REMEDY: Resolve the error in the RBAF and manually rerun the report program.

File parameter must be specified; PROGRAM ABORTED

CAUSE: The FILE parameter was not passed to the report program in the startup message.
REMEDY: Check the RIP processing to ensure that the FILE parameter is passed correctly and then rerun the report program manually.

Invalid file param value: **********; PROGRAM ABORTED

WHERE: '**********' specifies the invalid value of the FILE param.
CAUSE: An invalid FILE param was passed to the report program in the startup message.
REMEDY: Check the RIP processing to ensure that the FILE param is passed correctly and then rerun the report program manually.

Invalid parameter **********; PROGRAM ABORTED

WHERE: '**********' specifies the invalid value of the param.
CAUSE: The invalid param was passed to the report program in the startup message.
REMEDY: Check the RIP processing to ensure that the startup message is correct and then rerun the report program manually.

MTAF file empty; REPORT ABORTED

CAUSE: The indicated MTAF to be analyzed contains no records.
REMEDY: Check that the correct MTAF was being analyzed and the LCONF assign for the MTAF is correct.

RBAF file empty; REPORT ABORTED

CAUSE: The indicated RBAF to be analyzed contains no records.
REMEDY: Check that the correct RBAF was being analyzed and the LCONF assign for the RBAF is correct.

MTAF file empty; REPORT ABORTED

CAUSE: The indicated MTAF to be analyzed contains no records.
REMEDY: Check that the correct MTAF was being analyzed and the LCONF assign for the MTAF is correct.

WARNING Mismatch between MTAF Records (#) and days in Month (#)

WHERE: '#' specifies the number of MTAF records.
'#' specifies the number of days in the month.
CAUSE: The number of MTAF records being analyzed does not agree with the number of days in the month for which the MTAF was created.
REMEDY: Check the number of records in the MTAF being analyzed. If necessary, rebuild the MTAF with the correct number of records and manually rerun the report program.

WARNING MTAF CONTAINS (#) RECORDS WITH AN OLD VERSION NUMBER

WHERE: '#' specifies the number of MTAF records.
CAUSE: MTAF Records were created using an old version of the daily report program.
REMEDY: Manually rebuild the MTAF using the latest version of the daily report program.

WARNING Mismatch between RBAF Records (#) and days in Month (#)

WHERE: '#' specifies the number of RBAF records.
'#' specifies the number of days in the month.
CAUSE: The number of RBAF records being analyzed does not agree with the number of days in the month for which the RBAF was created.
REMEDY: Check the number of records in the RBAF being analyzed. If necessary, rebuild the RBAF with the correct number of records and manually rerun the report program.

WARNING RBAF CONTAINS (#) RECORDS WITH AN OLD VERSION NUMBER

WHERE: '#' specifies the number of RBAF records.
CAUSE: RBAF Records were created using an old version of the daily report program.
REMEDY: Manually rebuild the RBAF using the latest version of the daily report program.

WARNING Param LOGICAL-NET not found in LCONF

CAUSE: The LCONF LOGICAL-NET param is missing and, therefore, the Logical Network is not displayed correctly on the report output.
REMEDY: If it is required that the Logical Network is shown correctly on the report output, ensure that the LCONF LOGICAL-NET param is added to the LCONF. Rerun the report program manually.